

PALKEM, Sweden

WMO: 021640

Lat: 66.383N

Lon: 21.633E

Elev: 863

StdP: 14.24

Time Zone: 1.00 (EUC)

Period: 83-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-10.2	-19.5	2.0	-15.3	-14.0	2.7	-10.0	26.8	31.2	23.1	28.5	2.0	320	0.736

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.1	(c) 72.0	(d) 58.0	(e) 69.0	(f) 56.6	(g) 65.7	(h) 55.2	(i) 60.1	(j) 67.8	(k) 58.3	(l) 66.1	(m) 56.6	(n) 63.4	(o) 7.6	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.1	71.8	62.1	55.2	66.9	60.5	53.4	62.6	58.9	26.9	68.0	25.6	66.4	24.6	63.5	66.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	21.2	16.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-21.7	63.3	8.6	2.5	-28.0	65.1	-33.0	66.5	-37.9	67.9	-44.1	69.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.0	12.4	15.0	21.0	30.4	42.6	51.8	55.6	52.0	42.8	32.7	21.4	16.9
	DBStd	17.31	12.68	12.47	8.22	6.85	7.16	6.40	4.61	5.78	5.84	8.61	9.15	10.97
	HDD50	6641	1166	980	899	588	256	51	7	46	225	538	858	1027
	HDD65	11686	1631	1400	1364	1038	694	397	291	402	666	1003	1308	1492
	CDD50	432	0	0	0	0	27	106	181	109	9	0	0	0
	CDD65	2	0	0	0	0	0	2	0	0	0	0	0	0
	CDH74	43	0	0	0	0	4	31	5	3	0	0	0	0
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0
	WSAvg	5.8	5.4	6.2	6.0	5.5	6.0	6.6	5.9	5.6	5.2	6.4	5.3	5.8
	PrecAvg	23.0	1.7	1.3	1.3	1.1	1.4	2.2	2.8	2.6	2.4	2.0	2.2	1.9
Precipitation	PrecMax	30.6	3.0	2.6	2.9	2.7	2.8	4.1	4.8	5.7	8.2	3.8	4.3	6.1
	PrecMin	15.8	0.0	0.1	0.4	0.3	0.3	0.1	0.8	0.6	0.7	0.7	0.6	0.2
	PrecStd	4.2	0.7	0.7	0.7	0.6	0.6	1.0	1.1	1.4	1.6	0.8	1.0	1.1
	PrecStd	4.2	0.7	0.7	0.7	0.6	0.6	1.0	1.1	1.4	1.6	0.8	1.0	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	41.3	42.7	52.3	72.9	78.0	74.6	74.8	61.1	51.5	41.4	39.8
		MCWB	35.8	36.7	36.0	42.1	56.1	61.1	58.9	62.6	52.7	44.3	37.6	35.9
	2%	DB	35.5	37.2	38.7	46.7	68.2	71.8	71.2	69.0	57.0	48.6	37.9	35.9
		MCWB	31.5	33.0	33.4	38.5	54.0	58.0	57.4	59.5	50.4	43.8	35.2	32.8
	5%	DB	31.9	33.9	35.2	43.2	61.2	67.7	68.5	64.9	53.9	46.4	35.0	33.7
		MCWB	28.8	30.5	31.2	36.4	49.8	55.2	56.6	56.3	48.9	42.5	33.0	31.3
	10%	DB	28.9	31.0	32.4	40.8	55.2	63.4	65.6	62.2	51.3	44.3	33.1	31.8
		MCWB	27.1	29.0	29.6	34.8	46.0	52.4	55.3	55.0	47.3	41.3	31.7	30.3
	0.4%	WB	35.8	37.0	36.8	43.4	57.4	61.9	60.9	63.3	54.5	48.3	38.3	36.2
		MCDB	40.2	41.4	41.5	49.3	69.8	73.2	68.4	71.7	58.5	49.2	40.4	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.7	33.4	33.9	39.3	54.2	59.2	59.1	60.8	52.3	45.3	35.8	33.4
		MCDB	34.7	36.3	37.6	45.4	67.2	70.6	67.2	66.0	55.5	46.7	37.5	35.0
	5%	WB	29.3	31.4	31.7	37.2	50.4	56.5	57.9	58.4	50.0	43.2	33.4	31.8
		MCDB	31.4	33.4	34.4	41.9	60.6	65.8	65.5	63.5	52.2	46.2	34.7	33.2
	10%	WB	26.9	29.0	30.0	35.3	46.7	53.8	56.7	56.0	48.4	40.9	32.0	30.4
		MCDB	29.0	31.0	32.1	39.7	54.5	61.8	63.4	60.7	50.4	43.9	33.0	31.8
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.3	12.3	13.3	15.2	15.0	14.1	12.9	11.1	9.0	8.7	10.1
		MCDBR	10.2	9.7	12.7	15.7	24.2	22.3	19.4	17.0	14.0	10.6	8.0	11.3
		MCWBR	9.0	8.1	9.8	10.2	12.9	11.6	9.3	9.3	8.6	8.4	7.2	10.6
	5% WB	MDBR	10.3	9.1	10.7	13.8	23.2	19.9	16.2	14.0	10.4	9.8	6.7	9.7
		MCDBR	9.7	7.8	8.6	9.3	12.6	11.4	8.7	8.6	7.3	8.5	6.3	9.4
		MCWBR	9.7	7.8	8.6	9.3	12.6	11.4	8.7	8.6	7.3	8.5	6.3	9.4
Clear-Sky Solar Irradiance	taub		0.185	0.253	0.285	0.316	0.330	0.334	0.345	0.329	0.300	0.277	0.338	N/A
	taud		1.947	2.157	2.241	2.307	2.480	2.497	2.481	2.510	2.569	2.405	2.742	N/A
	Ebn at Noon		76	193	247	268	277	278	271	264	248	189	82	N/A
	Edh at Noon		5	16	25	30	29	29	29	25	19	13	5	N/A
All-Sky Solar Radiation	RadAvg		25	203	684	1278	1571	1655	1541	1139	638	232	48	3
	RadStd		2	29	53	62	158	129	149	102	92	39	6	0

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page